



Controlled urea release from CMC-based composites: effects of citric acid crosslinking and biochar incorporation

Amanda S. Giroto¹ · Thaynara R. Prado¹ · Amanda Yashima¹ · Stella F. Valle² · Bruna Lendzion Alves³ · Maraisa Gonçalves¹

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Abstract

Improving nitrogen (N) use efficiency while reducing environmental losses remains a major challenge in agriculture. Here, we show that a single polymeric matrix based on carboxymethyl cellulose (CMC) can be effectively engineered to control urea release when structurally tailored with citric acid as a crosslinking agent and biochar as a physical barrier. The formulation containing 10 wt % citric acid (CMC/U/BC/CA 10) exhibited the best performance, reaching 7500% swelling and maintaining 60% of water retention after 120 h, while providing a significantly delayed release profile (~18% after 4 h and 80% at the end of the experiment), whereas pure urea dissolved rapidly. Higher CA levels were associated with lower swelling and greater matrix rigidity and morphological heterogeneity, which may favor preferential diffusion pathways and faster nutrient release (up to 90%). Kinetic analysis indicated that urea release involved an important diffusional contribution coupled with polymer-chain relaxation and formulation-dependent structural changes, rather than a purely Fickian transport mechanism. Under incubation conditions, the composites reduced measured N-NH₃ volatilization relative to pure urea and affected exchangeable N-NH₄⁺ dynamics. The study demonstrates that a single biodegradable polymer matrix can efficiently regulate nutrient release, providing a simple, scalable, and sustainable strategy for the development of advanced nitrogen fertilizers.

Keywords Controlled-release fertilizer · Carboxymethyl cellulose · Urea · Biochar · Nitrogen use efficiency · Ammonia volatilization



Introduction

The expansion of the global population demands high food production, making agricultural practices fundamental to securing this supply. Nitrogen (N) inputs are indispensable for maintaining productivity levels in contemporary agriculture. However, low N use efficiency (NUE) remains a limiting factor in cropping systems due to the insufficient recovery rate of N fertilizers, which frequently ranges between 30% and 50% [1, 2]. Urea (U) stands out globally as the most consumed N fertilizer due to its high N concentration (46%), low operational cost, high solubility, and ease of management [3]. Nevertheless, urea's rapid dissolution results in significant N losses through ammonia (NH_3) volatilization, nitrate (NO_3^-) leaching, denitrification, and nitrous oxide (N_2O) emissions, drastically reducing the NUE [4]. It is estimated that up to 60% of the applied fertilizer is lost to the environment, inducing impacts such as soil acidification and compaction, atmospheric pollution, and the eutrophication of aquatic ecosystems [5, 6]. In this scenario, the careful control of fertilizer application, using minimal yet effective quantities, constitutes a fundamental strategy to mitigate these effects, contributing to the preservation of soil nutrients and to the enhancement of agricultural productivity from a sustainable perspective [7].

Controlled-Release Fertilizers (CRFs) have emerged as an effective approach to better synchronize nutrient availability with plant demand, reducing nutrient waste and significantly improving NUE. These products can reduce fertilizer application by 20% to 30% without compromising productivity [8, 9]. Polymeric matrices are frequently used as physical barriers to control the diffusion and release rate of nutrients [10–12]. However, many commercial products are still based on non-biodegradable synthetic materials [13].

Carboxymethyl cellulose (CMC) stands out as a promising matrix for CRFs due to its biodegradability, abundance, low toxicity, renewable origin, high hydrophilicity, and ability to form films and hydrogels [14–16]. However, the high water affinity of CMC may also represent a limitation, since insufficient stabilization of the polymeric network can lead to excessive swelling or even dissolution, causing nutrient release to be governed by matrix disintegration rather than by controlled diffusion [17]. Crosslinking is therefore an important strategy for regulating the water stability and transport properties of CMC-based matrices. Among the available crosslinkers, citric acid (CA) is particularly attractive because it is a low-toxicity, renewable polycarboxylic acid that promotes esterification with the hydroxyl groups of cellulose derivatives [18].

Several strategies involving CMC have consequently been investigated for urea delivery, including urea-loaded CMC hydrogels, CMC-containing composite films, and polymer-coated urea systems. Citric acid crosslinking has been successfully applied to CMC- and cellulose-based matrices for controlled nutrient delivery [19–21], including CA-crosslinked CMC hydrogels specifically designed for gradual urea release [19], while CMC-containing multicomponent coatings have also been used as diffusion barriers around conventional urea granules [22]. More complex composite approaches have incorporated carbon-rich phases, such as artificial humus, into CMC/CA matrices to further modify water uptake and nutrient transport [23]. These studies demonstrate the versatility of CMC-based systems; however, their

architectures, crosslinking strategies, carbonaceous components, and nutrient-loading approaches differ substantially, making direct structure–release relationships difficult to establish. A systematic comparison of representative CMC-based controlled-release fertilizer systems, including their formulation architecture, crosslinking strategy, carbonaceous components, release performance, and main limitations, is provided in Table S1.

Recent developments in biodegradable polysaccharide-based hydrogels have increasingly focused on tailoring network architecture and crosslinking to regulate water uptake and nutrient transport, with nutrient release frequently governed by diffusion through the swollen polymer network [24]. In parallel, the incorporation of BC into polymeric and polysaccharide-based matrices has emerged as a promising strategy to combine the water-retention capacity of hydrogels with the adsorption capacity and porous structure of BC [25, 26]. Recent BC-based systems have also demonstrated the potential of incorporating urea-loaded BC into biodegradable polymeric matrices to further regulate nutrient delivery [27].

Biochar (BC) represents an attractive carbonaceous component for this purpose because its heterogeneous structure and surface functional groups may provide additional interactions with N species and modify the diffusion pathways within the polymeric matrix [28]. In soil, BC may also contribute to nutrient retention, soil structure, and organic carbon management [29, 30]. Although BC has been investigated in controlled-release fertilizers [31, 32] and CMC has been used either as a fertilizer matrix or as a binder/coating component [19, 33, 34], the combined influence of BC incorporation and controlled CA crosslinking on the structure–property–release relationships of CMC/urea composites remains insufficiently understood. In particular, limited information is available on how systematic changes in CA content affect matrix morphology and stability, water uptake, urea diffusion, and the subsequent transformation and retention of released N under soil conditions.

Therefore, rather than simply introducing additional components into the formulation, the present study was designed to establish a direct relationship between matrix structure and fertilizer performance. A hybrid CMC/U/BC/CAC/CA composite was prepared in which U and BC were dispersed within the CMC matrix, while a non-crosslinked control and three CA levels (10, 20, and 30 wt% relative to CMC) were investigated, with the remaining formulation parameters maintained constant. The selected CA concentrations were based on previous studies demonstrating the strong influence of CA content on the structural and hydration properties of CMC-based networks. For instance, Valle et al. [35] reported a marked decrease in the swelling ratio of CMC hydrogel films when the CA concentration was increased from 5 to 10 wt%, while Nongnual et al. [36] demonstrated that CA contents within the 5–20 wt% range significantly affected the crosslinking and water-related properties of CMC films. Based on these findings, the present study included a non-crosslinked control and CA levels of 10 and 20 wt%, within the range previously investigated for CMC-based matrices, together with a higher level of 30 wt% to evaluate whether further increasing CA availability would enhance matrix stabilization or adversely affect the balance between network stability and hydration.

We hypothesized that CA-mediated crosslinking would regulate the hydration and diffusion properties of the CMC matrix, with an intermediate crosslinking level

providing a favorable balance between swelling and structural stability and, consequently, slowing urea diffusion. In addition, BC incorporation was hypothesized to increase the tortuosity of diffusion pathways and provide additional interactions with N species, potentially contributing to modifying N transformations and reducing volatilization losses.

Based on these hypotheses, the present study aimed to systematically investigate the effects of CA crosslinking and BC incorporation on the structural and functional properties of CMC-based urea composites. This experimental design enabled the effect of crosslinking level to be correlated with microstructure, thermal behavior, swelling, water retention, urea-release kinetics, NH_3 volatilization, and NH_4^+ formation. In this way, the study provides a systematic structure–property–release assessment of a CMC-based hybrid fertilizer and identifies the formulation conditions that provide a favorable balance between matrix hydration, structural stability, and controlled urea release.

Materials and methods

Preparation of CMC-based composites

Carboxymethyl cellulose (CMC)-based composites were prepared using (Na-CMC, analytical grade, $\geq 99.5\%$ purity; CC09494RA), supplied by Êxodo Científica (Brazil); urea (U) ($\text{CH}_4\text{N}_2\text{O}$, 99.5% purity, 46% of N and molar mass 60.06 g mol^{-1}), supplied by Neon (Brazil); citric acid (CA; $\text{C}_6\text{H}_8\text{O}_7$, $\geq 98\%$ purity, molar mass $192.12 \text{ g mol}^{-1}$), supplied by Dinâmica (Brazil); and commercial bamboo-derived biochar (BC), supplied by Takuatéc Florestal (Brazil). According to the manufacturer's specifications, the Na-CMC has a viscosity $\geq 2,000 \text{ mPas}^{-1}$, a pH of 5–9 (0.1% aqueous solution), and a density of 1.59 g cm^{-3} . The BC exhibited a BET specific surface area $< 10 \text{ m}^2 \text{ g}^{-1}$, as determined by N_2 adsorption analysis performed previously on the same material batch.

For composite preparation, an aqueous CMC solution (2 wt%) was first prepared by dispersing the polymer in deionized water under continuous magnetic stirring at 150 rpm and 25°C for 2 h, until complete dissolution and formation of a homogeneous viscous solution. U was then added at a mass ratio corresponding to 50 wt% relative to the CMC content (CMC/U system), and the mixture was stirred at 150 rpm for 30 min to ensure uniform dispersion and interaction between U and the polymer matrix. Immediately after U incorporation, BC was added at a fixed content of 1 wt% relative to the total composite mass, and the mixture was maintained under stirring at 150 rpm to promote homogeneous distribution of the BC particles within the polymer–U matrix. CA was subsequently added as a crosslinking agent at different concentrations (0, 10, 20, and 30 wt%, relative to the CMC mass) to obtain the different formulations. After CA addition, the mixtures were homogenized under stirring at 150 rpm, and the temperature was gradually increased to 80°C to promote CA-mediated crosslinking of the CMC matrix. The systems were maintained under constant stirring at 150 rpm and 80°C for 2 h. Following this step, the resulting viscous mixtures were transferred to polypropylene Petri dishes and dried in a forced-air

oven at 50 °C for 12 h. After drying, the self-supporting composite films were cut into circular pieces with a diameter of 5 mm and stored in a desiccator at room temperature (25 °C) until further characterization and testing. The resulting composites were denoted as CMC/U, CMC/U/BC/CA0, CMC/U/BC/CA10, CMC/U/BC/CA20, and CMC/U/BC/CA30. A schematic illustration of the composite preparation is shown in Fig. 1.

Composites characterizations

Composites were characterized by X-ray powder diffraction (Rigaku Ultima IV, Rigaku Corporation, Japan) using Cu K α radiation ($\lambda=0.154$ nm) at 40 kV and 30 mA. Data were acquired at $2\theta=20^\circ \text{ min}^{-1}$ from 10 to 80°. Electron microscopy (SEM) was performed on an FEI Inspect TM S50 V (FEI Company, USA). Samples were sputter-coated (Quorum Tech., Q150R, Quorum Technologies Ltd., United Kingdom) with a gold film, and then analysis was performed at an operating voltage of 10 kV. Fourier transform infrared (FTIR) spectroscopy was done using a Lyza 7000 FTIR spectrometer (Anton Paar GmbH, Austria) equipped with an attenuated total reflectance (ATR) accessory. Dried samples were analyzed directly on the ATR crystal, and spectra were recorded in the range of 4000–400 cm^{-1} with a spectral resolution of 4 cm^{-1} , accumulating 32 scans per sample. All spectra were baseline-corrected and normalized when required for comparison. Thermal stability of the composites was evaluated in a thermogravimetric analysis (TGA) using a TA Instruments Q5000 (TA Instruments, USA). Approximately 5.0 mg of each dried composite was placed in an alumina crucible and heated from room temperature to 600 °C at a heating rate of 10 °C min^{-1} under a nitrogen atmosphere (purge flow: 50 mL min^{-1}).

Swelling in water

The swelling capacity of the composites was evaluated by immersing a known mass of the dry composite in water at 25 °C. After 15, 30, 60, and 90 min of immersion, the



Fig. 1 Schematic illustration of the preparation of CMC-based composites containing urea (U), biochar (BC), and different citric acid (CA) contents (0, 10, 20, and 30 wt%)

composites were removed, gently blotted with filter paper to eliminate excess surface moisture, and weighed. The swelling degree was calculated using Eq. 1:

$$SR\% = \frac{(m_t - m_0)}{m_0} \times 100 \quad (1)$$

where m_0 is the initial mass of the dry sample (g), and m_t is the mass of the swollen sample (g) at time.

Water retention assessment in soil

For the water-retention assay, sandy soil was air-dried and sieved through a 2 mm mesh to ensure uniform particle size. A portion of composite samples (0.1 g) was thoroughly blended with 20 g of soil in a 100 mL plastic tube. A control tube containing only 20 g of soil was also prepared. Subsequently, 10 mL of deionized water was added to each tube, and then the tubes were placed in a temperature-controlled oven set at 40 °C. At 24 h intervals, the tubes were removed and weighed again to obtain the mass. All measurements were performed in triplicate. The soil water-retention rate (WR) was calculated according to Eq. 2:

$$WR\% = \frac{m_t - m_0}{m_s - m_0} \times 100 \quad (2)$$

where m_0 is the initial dry mass of the sample (g), m_s is the mass of the swollen sample at the beginning of the water-retention experiment (g), and m_t is the sample mass (g) measured at time.

Urea release in aqueous medium

Each composite tested for urea release contained the same initial total N content. The release rate at 25 °C was evaluated by placing the films into 250 mL beaker flasks with water and gently agitating them for 8 days using an orbital shaker set to 120 rpm (Ethik Technology, Brazil). Commercial urea granules were included as a control. Urea solubilization and release kinetics were quantified in triplicate following the methodologies described by [37] and [38].

This aqueous immersion test was designed as an accelerated assessment of the ability of the composite matrices to retard urea dissolution under conditions of direct and continuous contact with water, rather than to reproduce the long-term release period expected under field conditions.

Kinetics of U release

The release kinetics of U from the composites were analyzed using zero-order, first-order, Higuchi, and Korsmeyer–Peppas models. The zero-order model describes a constant release rate independent of concentration and is expressed as a linear Eq. 3:

$$C_t = C_0 + k_0 t \quad (3)$$

The first-order model assumes that the release rate depends on the remaining concentration in the matrix, following Eq. 4:

$$\log C_t = \log C_0 - \frac{k_1 t}{2.303} \quad (4)$$

The Higuchi model is based on diffusion-controlled release and is given by Eq. 5:

$$C_t = k_H t^{1/2} \quad (5)$$

The Korsmeyer–Peppas model, a semi-empirical approach used to elucidate the release mechanism, is described by Eq. 6:

$$\frac{C_t}{C_{eq}} = k_{KP} t^n \quad (6)$$

where n is the diffusion exponent. In these equations, C_t represents the cumulative amount of urea released at time t , C_{eq} is the total amount released at equilibrium, and k_0 , k_1 , k_H and k_{KP} are the kinetic constants of the respective models [34, 39–41].

The Peppas–Sahlin model was additionally applied to distinguish the contributions associated with molecular diffusion and polymer-chain relaxation:

$$\frac{C_t}{C_\infty} = k_1 t^m + k_2 t^{2m} \quad (7)$$

where k_1 and k_2 are the kinetic constants associated with the diffusional and polymer-relaxational contributions, respectively, and m is the diffusional exponent. For the Peppas–Sahlin model, the mean release data from the three replicates were normalized as C_t/C_∞ , with C_∞ estimated from the equilibrium region (48–54 h), and the initial release data up to 480 min were fitted by nonlinear regression. The coefficients k_1 and k_2 were constrained to non-negative values. Pure U was not fitted to the Peppas–Sahlin model because it does not contain a polymeric matrix. In all equations, C_t represents the cumulative amount of U released at time t , C_∞ is the amount released at equilibrium, and the corresponding kinetic constants are defined according to each model [42].

The time-dependent contributions predicted by the Peppas–Sahlin model were calculated from the diffusional $F = k_1 t^m$ and $R = k_2 t^{2m}$ relaxational terms. Their relative contributions to the modeled release at a given time were calculated as:

$$F (\%) = \frac{k_1 t^m}{k_1 t^m + k_2 t^{2m}} \times 100 \quad (8)$$

and

$$R(\%) = \frac{k_2 t^{2m}}{k_1 t^m + k_2 t^{2m}} \times 100 \quad (9)$$

These contributions were used to assess changes in the relative importance of diffusion and polymer relaxation during the initial release period.

N release and transformation in soils

The N release from the CMC/U/BC/CA composites and urea was evaluated by incubation in sand. Samples were incorporated into the sand at a 1000:1 g g⁻¹ sand: N ratio in 125 mL polyethylene screw-cap bottles and moistened with deionized water to 80% of the water-holding capacity (WHC). A container containing 5 mL of 4% boric acid (Sigma-Aldrich) was placed inside each bottle to capture NH₃ volatilized during incubation. Independent experimental units were prepared for each incubation time and maintained at 40 °C. At each sampling time, the corresponding bottles were removed from the experiment and analyzed. Therefore, N-NH₃ measured at each sampling time represents the cumulative amount volatilized from the respective experimental unit up to that time, rather than sequential measurements from the same bottle. Exchangeable N-NH₄⁺ was likewise determined using independent experimental units collected at each sampling time. N-NH₃ volatilization was evaluated after 3, 7, 14, 28, and 42 days, whereas exchangeable N-NH₄⁺ was determined after 3, 7, 15, 30, and 42 days.

The N transformations were evaluated based on the methodologies described by Giroto et al. and Guimarães et al. [43, 44], considering cumulative NH₃ volatilization and exchangeable NH₄⁺ recovered from the sand. The NH₃ volatilization was determined by titration of boric acid with HCl (0.01 mol L⁻¹, Synth, Brazil) in Erlenmeyer Flasks, and soil N was extracted by KCl 1 mol L⁻¹ (Synth, Brazil) in a 10:1 ratio (extractant: soil) by stirring the solution in an orbital shaker (Ethik Technology, Brazil). To inactivate the urease enzyme, a solution of 5 mg L⁻¹ of phenyl-mercuric acetate (Sigma-Aldrich, USA) was added at the extraction stage [45]. The suspension was stirred for 1 h, filtered, and stored in a refrigerator (5 °C). Exchangeable N-NH₄⁺ was extracted from the sand and analyzed colorimetrically [46]. The contents recovered in each N fraction were expressed as percentages of the total N. The mean values and standard deviation were recorded for partial N recovery as volatilized NH₃ and NH₄⁺ formation in the soil.

At the end of the 42-day incubation period, partial N recovery was calculated considering the two N pools quantified in the experiment: cumulative volatilized N-NH₃ and exchangeable N-NH₄⁺. For each experimental replicate, measured partial N recovery was calculated as:

$$N_{\text{recovered}} = N_{\text{NH}_3} + N_{\text{NH}_4^+}$$

Considering the initial application of 10 mg N per experimental unit, partial N recovery was expressed as a percentage of the applied N according to:

$$N_{\text{recovery}}(\%) = \frac{N_{\text{NH}_3} + N_{\text{NH}_4^+}}{10} \times 100$$

The fraction of N not recovered in these two measured pools was calculated as:

$$N_{\text{unaccounted}} = 10 - (N_{\text{NH}_3} + N_{\text{NH}_4^+})$$

Because N-NO_3^- and residual matrix-associated N were not directly quantified in the present experiment, this fraction was conservatively classified as unaccounted N and was not assigned to any specific N pool. Accordingly, the analysis presented here represents a partial N recovery based on the measured N-NH_3 and N-NH_4^+ pools rather than a complete N balance.

Differences among treatments in cumulative N volatilization as NH_3 and exchangeable NH_4^+ at the end of the aerobic incubation period (42 days) were assessed by one-way analysis of variance (ANOVA). Before ANOVA, normality of residuals and homogeneity of variances were evaluated using the Shapiro–Wilk and Levene tests, respectively (Table S5, Supplementary Material). Exchangeable N-NH_4^+ data were log-transformed before statistical analysis to meet the normality assumption. When the ANOVA F-test was significant, treatment means were compared using Tukey's HSD test at $p < 0.05$. All statistical analyses were performed using OriginPro 2018 (OriginLab Corporation, Northampton, MA, USA). Results are presented in their original units for ease of interpretation.

Results and discussion

Figure 2 evidenced clear morphological differences among the samples. Pure U exhibited well-defined prismatic crystals completely exposed at the surface [38, 47]. In CMC/U, these crystals appeared partially covered by or embedded within a continuous polymeric matrix, suggesting a close physical association between urea and CMC. However, the SEM observations alone are not sufficient to confirm physical encapsulation of urea within the polymeric matrix. The material without CA showed a smoother surface with larger, more rectangular, and still well-defined urea crystals. In contrast, the addition of BC resulted in a rougher and more heterogeneous morphology, with larger and less ordered crystalline features. In some regions, porous features associated with the BC structure could also be observed. Considering the BC-containing composites, CMC/U/BC/CA10 exhibited a relatively cohesive morphology, although some surface heterogeneity remained visible. In contrast, CMC/U/BC/CA20 showed the most compact and homogeneous microstructure among the evaluated formulations, with a more continuous surface and fewer apparent morphological irregularities. CMC/U/BC/CA30 exhibited a more granular and fragmented morphology. These observations indicate that CA content influenced the organization of the composite matrix [39, 48]; however, the morphological differences observed by SEM alone cannot be directly attributed to differences in crosslinking density.

Figure 3 presents the X-ray diffraction patterns of the composites. The main diffraction peaks observed in the urea-containing samples are consistent with crystalline

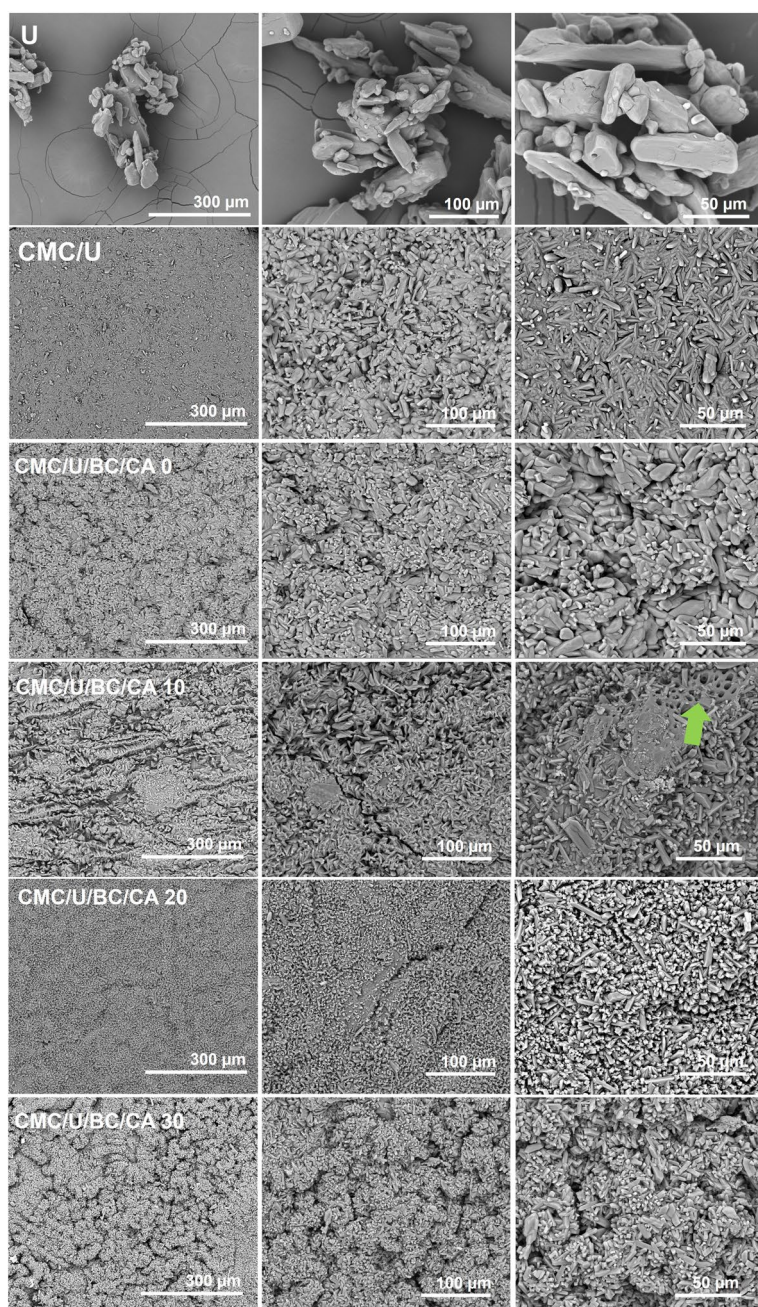


Fig. 2 SEM micrographs of pure urea (U) and the developed composites (CMC/U, CMC/U/BC/CA0, CMC/U/BC/CA10, CMC/U/BC/CA20, and CMC/U/BC/CA30), obtained at magnifications of 400 $\times$, 800 $\times$, and 1500 $\times$ (from left to right), respectively

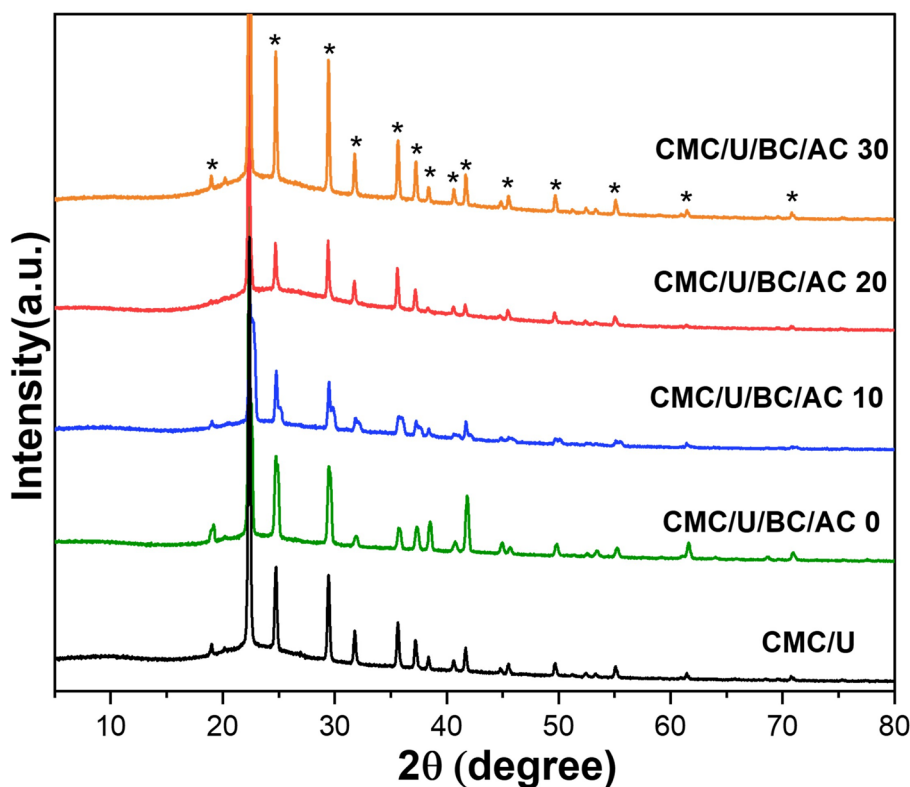


Fig. 3 X-ray diffraction (XRD) patterns of pure urea and CMC-based composites containing different CA contents. Asterisks (*) indicate diffraction peaks associated with crystalline urea according to the ICSD reference pattern (ICSD No. 01-075-0653)

urea according to the reference pattern from the ICSD database (ICSD No. 01-075-0653). Interestingly, the CMC/U/BC/CA10 sample exhibited marked changes in the diffraction profile, including the appearance of additional reflections and redistribution of the relative peak intensities compared with pure urea and the other formulations. These changes indicate that the incorporation of 10 wt% CA modifies the crystalline organization of urea within the composite matrix. However, XRD alone does not allow unambiguous assignment of these reflections to a specific urea–citrate molecular phase, co-crystal, or eutectic-derived polymorph. Interactions between urea and citric acid, including the formation of urea–citric acid structures under specific reaction conditions, have been previously reported [47]. Nevertheless, the diffraction changes observed here cannot be considered sufficient evidence for the formation of an analogous phase within the CMC-based composite. At higher CA contents (20–30 wt%), the diffraction profiles showed a greater contribution of reflections associated with the individual crystalline components, together with changes in peak intensity and definition, indicating further modification of the crystalline/amorphous organization of the composite.

The FTIR spectra of the precursor materials (Fig. 4A) and composites (Fig. 4B) show that characteristic absorption bands of the starting components are retained after composite formation. Pure urea exhibited prominent bands at approximately 1679, 1590, and 1459 cm^{-1} . In the composites, the band at 1676 cm^{-1} is mainly attributed to the C=O stretching vibration of urea, consistent with values around 1675–

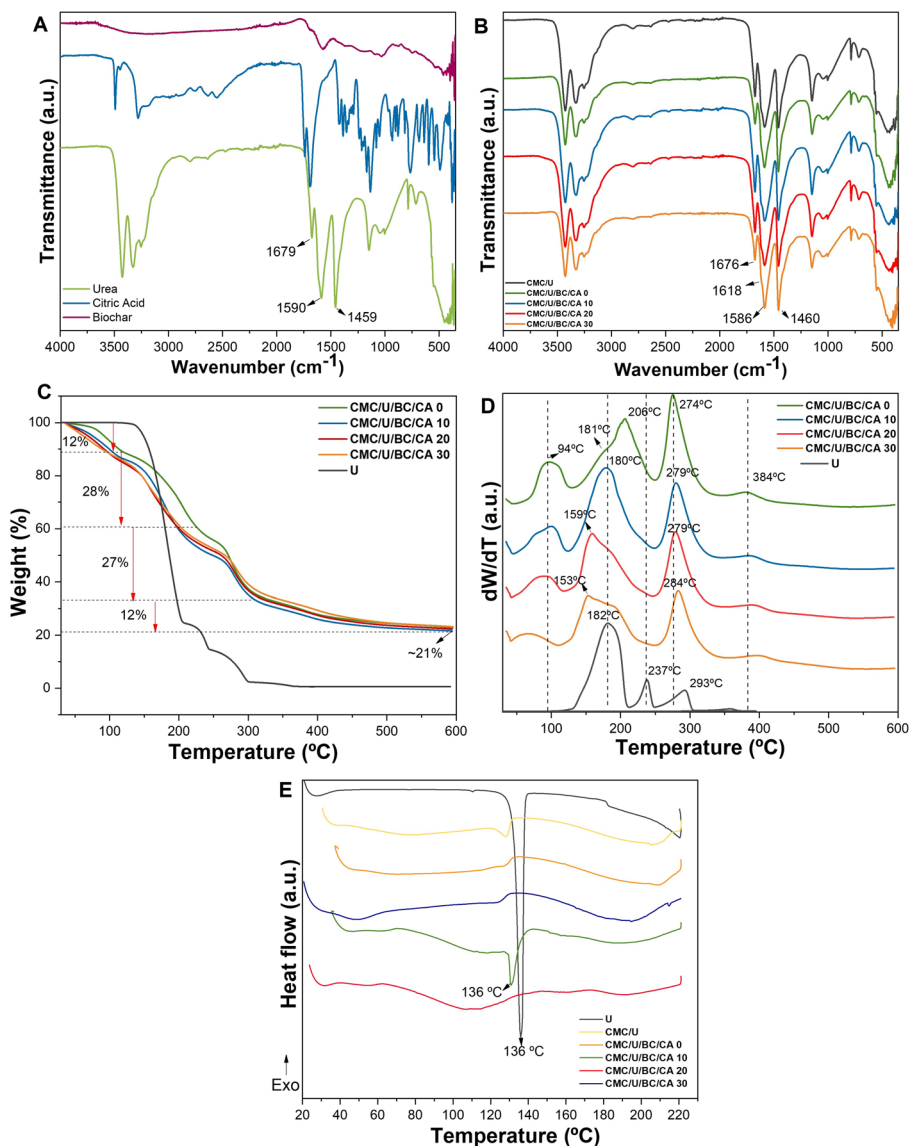


Fig. 4 Structural and thermal characterization of the starting materials and CMC-based composites: (A) FTIR spectra of the precursors, (B) FTIR spectra of the composites, (C) thermogravimetric (TG) curves, (D) derivative thermogravimetric (DTG) curves, and DSC curves of pure urea and the resulting composites, showing formulation-dependent changes in the thermal event associated with urea

1677 cm^{-1} reported for urea-loaded CMC-based matrices [39, 49]. The bands at 1618 and 1586 cm^{-1} may contain overlapping contributions from N–H vibrations of urea and the asymmetric stretching of carboxylate groups (COO^-) of CMC [39, 40, 50]. Similarly, the band around 1460 cm^{-1} may contain contributions from the symmetric stretching of COO^- groups of CMC and C–N/N–H vibrations of urea [14, 40]. Changes in the position and relative intensity of these bands, together with modifications in the broad O–H/N–H region, indicate changes in the local environment of the functional groups after composite formation and are consistent with intermolecular interactions among the components. However, because CMC, urea, and CA vibrations substantially overlap, these spectral changes cannot be unambiguously assigned to a specific interaction or considered, by themselves, evidence of a new chemical phase.

The thermal behavior of the composites was investigated by TGA/DTG analysis, as shown in Fig. 4C and D. The first thermal event observed around 94 °C corresponds to the evaporation of physically adsorbed water and volatile molecules from the hydrophilic polymeric matrix (12% weight loss). A second decomposition stage is observed between approximately 150 and 200 °C, which is mainly associated with the thermal decomposition of urea (28% weight loss) [14]. Pure U exhibits a characteristic DTG peak at 182 °C, corresponding to its primary decomposition and formation of biuret through ammonia release [29]. In the composite, this peak shows slight shifts depending on the formulation, suggesting interactions between urea molecules and the surrounding polymer matrix. An additional decomposition event around 206 °C can be attributed to the degradation of intermediate products formed during urea decomposition, such as biuret and related nitrogen-containing compounds [48]. A major degradation stage occurs in the range of 270–285 °C (27% weight loss), which is attributed to the thermal decomposition of the CMC backbone. Notably, a gradual shift of this peak toward higher temperatures is observed with increasing CA content. The CMC/U/BC/CA 30 sample exhibits the highest decomposition temperature (~284 °C), indicating enhanced thermal stability of the composite. This behavior is consistent with increased stabilization of the polymeric network associated with CA-mediated crosslinking and the presence of BC. At higher temperatures, a broad decomposition event around 384 °C is observed (12% weight loss), which may be associated with the degradation of carbonaceous residues and the B structure. These results indicate that the composite structure increases the thermal stability of the polymeric matrix while maintaining the presence of U within the system.

The DSC curves (Fig. 4E) provided complementary evidence of changes in the thermal environment of urea after incorporation into the CMC-based matrix. Pure urea exhibited a sharp endothermic event at approximately 136 °C, characteristic of its melting behavior. In contrast, incorporation into the composites substantially modified this thermal event, with broadening, reduction, or suppression of the characteristic urea peak depending on the formulation. Similar modifications in the thermal behavior of CMC-based matrices after incorporation of urea and/or crosslinking have been reported for cellulose-derived fertilizer hydrogels [51–53]. These changes are consistent with alterations in the physical state and local environment of urea resulting from its dispersion and interactions within the polymeric network. Notably, CMC/U/BC/CA20 retained a comparatively well-defined endothermic event at

approximately 131 °C, although shifted to a lower temperature relative to pure urea. This behavior suggests that a fraction of urea remained in a more thermally ordered state in this formulation, whereas the broader or less-defined events observed for the other composites indicate a greater perturbation of its thermal behavior. Nevertheless, these DSC changes are not considered evidence of the formation of a specific urea–citrate co-crystal or eutectic-derived polymorph. Rather, when considered together with the XRD and FTIR results, they support formulation-dependent changes in the crystalline organization and local environment of urea within the composite matrix.

The swelling results (Fig. 5A) combined with the visual evolution of the composites during water immersion (Fig. 5B) reveal clear differences in hydration behavior and structural stability as a function of CA content.

The CMC/U and CMC/U/BC/CA0 systems showed limited measurable swelling and rapidly lost their structural integrity during immersion, forming a gelatinous, non-self-supporting phase. Consequently, the samples could not be reliably recovered for weighing at longer immersion times, and their swelling degree could not be determined. This behavior indicates that, in the absence of CA-mediated crosslinking, hydration of the CMC matrix was accompanied by loss of physical integrity, and the presence of urea and BC alone was insufficient to stabilize the polymer network in aqueous media. In contrast, the CA-containing composites maintained their physical

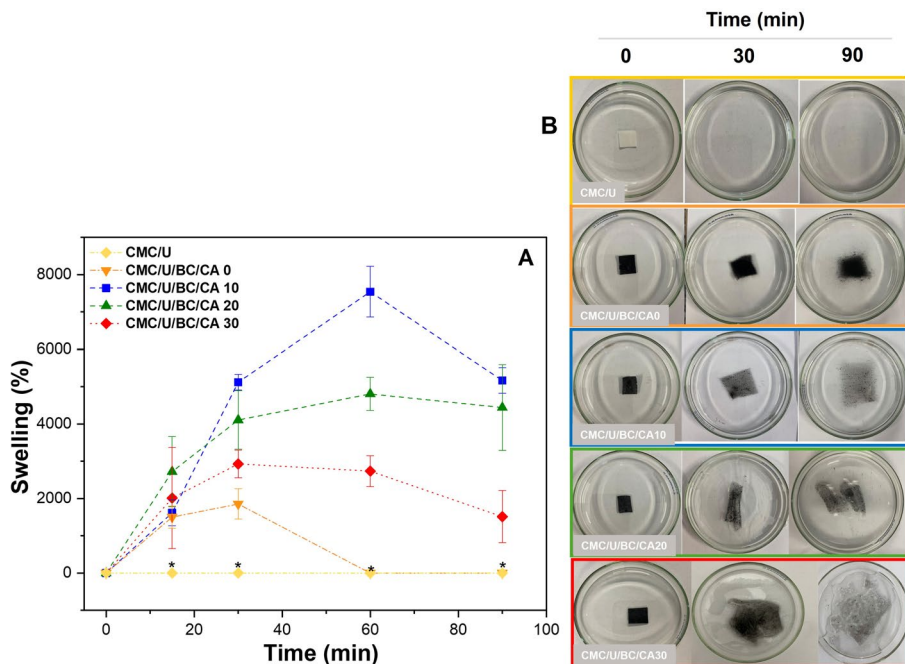


Fig. 5 Swelling capacity of CMC-based composites (A) and representative images of the samples after immersion for 0, 30, and 90 min (B). Values are expressed as mean $\pm$ standard deviation ($n=3$). The asterisk (*) indicates conditions in which the swelling degree could not be determined because the material either solubilized or lost its structural integrity, forming a viscous, non-self-supporting phase that could not be recovered for weighing. Statistical comparisons among formulations are provided in Table S2

integrity throughout the swelling experiment. CMC/U/BC/CA10 showed the highest swelling, reaching approximately 7500% after 60 min, compared with approximately 4800 and 2800% for CMC/U/BC/CA20 and CMC/U/BC/CA30, respectively. Statistical analysis confirmed significant differences among the crosslinked formulations at 60 min ($p < 0.05$), with CMC/U/BC/CA10 exhibiting significantly higher swelling than CMC/U/BC/CA20 and CMC/U/BC/CA30 (Table S2, Supplementary Material). At 90 min, CMC/U/BC/CA10 and CMC/U/BC/CA20 showed statistically similar swelling values, whereas CMC/U/BC/CA30 remained significantly lower. These results demonstrate that CA content significantly influenced the hydration behavior of the CMC-based matrices.

The swelling performance observed in this study is within the broad range reported for CMC-based materials developed for water and fertilizer management. Villalva-Cañavi et al. [53] reported a maximum swelling of approximately 1900% for citric-acid-crosslinked CMC hydrogels prepared in the presence of urea, while Laddawan et al. [12] obtained equilibrium swelling capacities of 7.6–10.3 g g⁻¹ for CMC hydrogels crosslinked with citric acid. Priya et al. [49] reported a substantially higher water absorption capacity of 147 g g⁻¹ for a urea-loaded cellulose nanofiber/CMC hydrogel, which was attributed to its porous morphology and abundance of hydrophilic groups. CMC-based hydrogels have also been successfully employed as matrices for urea loading and controlled fertilizer delivery, as demonstrated by Shang et al. [54]. These literature data illustrate the wide variation in swelling capacity achievable in CMC-based systems as a function of network composition, crosslinking chemistry, porosity, and experimental conditions. Additionally, these comparisons reinforce that swelling in CMC-based fertilizer matrices is governed not only by polymer hydrophilicity but also by the balance between network stabilization and the accessibility of water to the polymer chains.

Water retention is a key factor for plant growth and nutrient availability [55]. Statistical analysis of the data in Fig. 6 showed significant differences among treatments at 24 h ($p < 0.05$), whereas no significant differences were detected at 48, 72, 96, or 120 h (Table S3, Supplementary Material). The CMC/U sample initially showed improved water retention; however, this effect was not sustained over time. This behavior may be related to the rapid loss of hydrogel structure observed during the swelling experiment, which can temporarily promote water retention and aggregation between sand particles. As the structure progressively loses its integrity, the retention capacity decreases, approaching values similar to those of pure sand. At 120 h, CMC/U/BC/CA0 showed the highest mean residual water content ($59.0 \pm 3.3\%$), compared with $47.0 \pm 9.7\%$ for pure sand. Although this corresponds to approximately 25% higher residual water content, the difference was not statistically significant ($p > 0.05$). The CA-containing formulations showed intermediate values at this time, indicating that increasing CA content did not result in a progressive improvement in water retention. BC-containing matrices showed a tendency toward improved water retention at longer times, but these differences were not statistically significant at the end of the 120 h experiment.

Improved soil water retention has also been reported for different hydrogel-based fertilizer systems, although direct quantitative comparisons should be made cautiously because of differences in soil type, hydrogel loading, formulation, and experi-

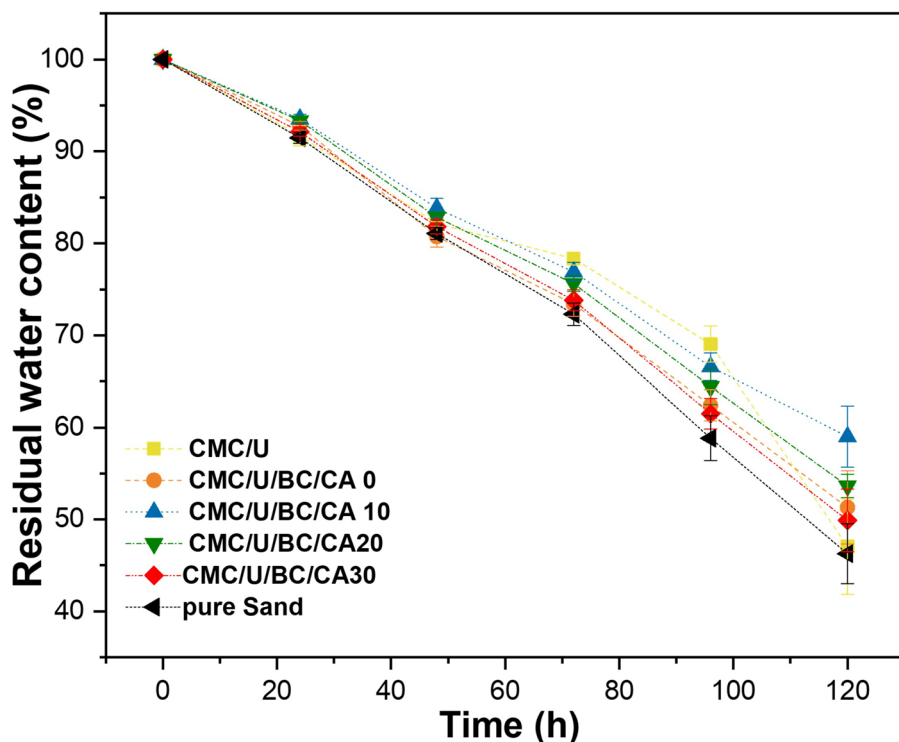


Fig. 6 Water retention of sand amended with the CMC-based composites as a function of time. Residual water content is expressed as a percentage of the initial water content. Statistical comparisons among treatments at each evaluation time are provided in Table S3

mental conditions. Cheng et al. [56] reported relative water contents of approximately 42–56% after 192 h for different soils amended with a water- and fertilizer-integrated hydrogel containing urea. Similarly, Das et al. [57] reported improvements in water-retention capacity of 57.20–67.61% using hydrogel–biochar composites designed for agricultural applications and controlled nutrient release. More closely related to the polymeric matrix used in the present study, CMC hydrogels derived from waste paper and sawdust increased the water-holding capacity of sandy soil from 25.5% to approximately 29.5–29.6% [19]. More recently, a multifunctional CMC-based hydrogel developed for simultaneous soil-moisture and fertilizer management increased soil water retention by approximately 37% at a hydrogel content of 5% (v/v) [58].

After evaluating the water absorption and water retention capacity of composites, urea release experiments were carried out in aqueous medium by immersing the composites in water at neutral pH and 25 °C.

Figure 7 shows that pristine urea dissolved almost instantaneously, with complete release occurring within the first minutes of water immersion. In contrast, all composite formulations markedly delayed urea diffusion; however, the mechanisms underlying this delay differ substantially depending on the structural stability of each system. Considering the accelerated nature of the aqueous immersion test, these results are interpreted as a comparative measure of the ability of the matrices to

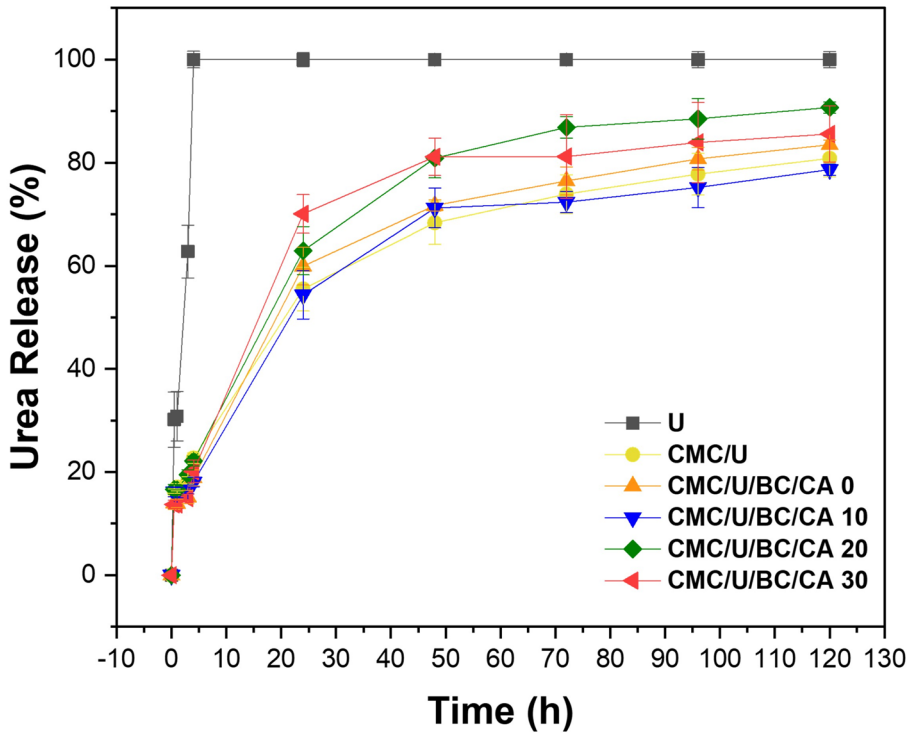


Fig. 7 Urea release profiles of CMC-based composites containing different CA contents compared with pristine urea during water immersion. Values are expressed as the percentage of urea released as a function of time

delay urea dissolution rather than as a simulation of long-term nutrient release under field conditions. The CMC/U/BC/CA10 system in particular displayed the steadiest release profile. While pure urea achieved full dissolution by the first 4 h, urea release from CMC/U/BC/CA10 was only 18% at this point. Although the release pattern shows an increase over the 24 h, it was still almost half of the amount from pure urea. Thereafter, CMC/U/BC/CA10 features a slower profile, reaching around 80% of U delivery by the end of the experiment, i.e., 20% lower than pure U. CMC/U/BC/CA30 released ~30% more urea than CMC/U/BC/CA10 and maintained higher values throughout the test. CMC/U/BC/CA20 features the highest urea release at 72 h, reaching the maximum of 90% at the end of the experiments.

Although increasing crosslinking generally reduces swelling and molecular diffusivity by restricting polymer-chain mobility, the release behavior observed here suggests a non-linear effect of CA content on the transport properties of the CMC/U/BC/CAC matrix. Previous studies on CA-crosslinked CMC systems have similarly demonstrated that variations in CA concentration strongly affect network organization, swelling, water resistance, and mechanical properties, and that increasing CA content does not necessarily result in a proportional improvement in the functional properties of the resulting matrix [36, 59, 60].

In the present system, CMC/U/BC/CA10 appears to provide the most favorable balance between network cohesion and hydration, resulting in a more continuous diffusion barrier. At higher CA contents (20 and 30 wt%), the greater restriction of polymer-chain mobility may increase matrix rigidity and alter its structural organization. These effects, together with the morphological heterogeneity observed by SEM, may affect the continuity and tortuosity of water and urea transport pathways. Therefore, the release behavior should not be interpreted as a direct increase in permeability caused by increasing crosslink density, but rather as a consequence of the non-linear interplay between crosslinking, matrix hydration, structural organization, and transport pathways.

The CMC/U, which does not contain B or a crosslinking agent, exhibited a rapid initial release compared to the other composites, associated with its highly hydrophilic nature and loosely structured polymeric network, allowing fast water uptake. However, this system also undergoes rapid structural collapse, as seen in the swelling capacity (Fig. 5), forming a viscous and gelatinous phase that acts as a secondary diffusion barrier, partially restricting urea release over time. As a result, its release becomes more comparable to that of the CMC/U/BC/CA10 formulation, although governed by a fundamentally different mechanism.

For this reason, several studies have explored the combination of CMC with other polymers or functional additives to improve urea release behavior, given the high solubility of both urea and CMC, which often leads to rapid dissolution and limited release control [61]. For example, CMC-based hydrogels combined with gelatin or chitosan have shown improved structural stability and prolonged nutrient release [55, 62], while PVA/CMC composite coatings have demonstrated reduced nutrient permeability and enhanced water retention [22]. These strategies highlight the importance of reinforcing the polymeric network to modulate diffusion and achieve more effective controlled-release behavior.

The release behavior observed in this study follows trends reported for biochar-based and CMC-containing controlled-release systems, where structured matrices significantly delay nutrient release compared to uncoated U. Conventional fertilizers typically release most of their nitrogen within the first day. At the same time, biochar-based systems can partially delay this process through adsorption interactions between nitrogen species and B functional groups, although still exhibiting a relatively rapid initial release [28, 32, 63]. In contrast, the incorporation of polymeric matrices further extends nutrient release, with cumulative values typically ranging from ~60–80% over 20–30 days, reflecting the role of the polymeric network as a diffusion barrier [22, 55]. More advanced composite systems, such as CMC/PVA-based coatings reinforced with functional additives, can further reduce nitrogen release to approximately 55–60% over 30 days and exhibit a two-stage release profile, characterized by an initial fast phase followed by a sustained diffusion-controlled regime [64].

Urea release data were further analyzed using zero-order, first-order, Higuchi, Ritger–Peppas, and Peppas–Sahlin models to evaluate the release kinetics and better distinguish the contributions associated with diffusion and polymer-chain relaxation. The fitted parameters are summarized in Table 1.

Table 1 Kinetic parameters obtained by fitting urea-release data to zero-order, first-order, Higuchi, and Ritger–Peppas models for CMC/U-based composites

Kinetic models									
Samples	Zero-order			First-order		Higuchi		Ritger–Peppas	
	k_0 (mg·L ⁻¹ ·min ⁻¹)	C_0	R^2	k_1 (min ⁻¹)	R^2	k_H (mg·L ⁻¹ ·min ^{-1/2})	R^2	k_{RP} (min ⁻ⁿ)	n
U	0.038	44.76	0.70	0.00014	0.67	2.060	0.98	0.1050	0.113
CMC/U	0.008	21.62	0.93	0.00019	0.85	0.806	0.99	0.0917	0.089
CMC/U/B 0	0.009	19.41	0.91	0.00022	0.83	0.89	0.98	0.0634	0.105
CMC/U/B 10	0.008	19.69	0.91	0.00021	0.863	0.81	0.97	0.1024	0.073
CMC/U/B 20	0.010	21.27	0.91	0.00021	0.837	0.96	0.97	0.0844	0.086
CMC/U/B 30	0.009	20.22	0.87	0.00023	0.803	0.964	0.933	0.0650	0.099
									0.88

All composites exhibited a good fit to the Higuchi model ($R^2=0.93\text{--}0.99$), indicating an important diffusional contribution to urea transport. Low release exponent (n) values were also obtained from the Ritger–Peppas model ($n<0.5$). However, considering the substantial swelling and structural changes observed during immersion, these parameters alone cannot be interpreted as evidence of a purely Fickian release mechanism. The Peppas–Sahlin model therefore provided a more appropriate framework for separating the diffusional and polymer-relaxational contributions to the initial release stage.

For CMC/U, CMC/U/BC/CA0, CMC/U/BC/CA20, and CMC/U/BC/CA30, positive values of both k_1 and k_2 were obtained, indicating that diffusion and polymer relaxation contributed simultaneously to urea release. The relative contributions calculated from the Peppas–Sahlin terms (Table S4) further showed that, for CMC/U, CMC/U/BC/CA0, CMC/U/BC/CA20, and CMC/U/BC/CA30, the diffusional contribution decreased from approximately 71–74% at 15 min to 61–63% at 480 min, while the relaxational contribution increased from approximately 26–29% to 37–39%. This evolution indicates that diffusion was an important contribution during the initial release period, while polymer relaxation became progressively more relevant as matrix hydration proceeded.

CMC/U/BC/CA10 showed a different fitting behavior, with k_2 approaching zero for the mean release profile. This result indicates that the diffusional and relaxational contributions could not be independently resolved by the Peppas–Sahlin model for this formulation and should not be interpreted as evidence of an exclusively Fickian mechanism. In fact, the pronounced swelling of CMC/U/BC/CA10 ($\sim 7500\%$) demonstrates substantial hydration and rearrangement of the polymeric network during immersion. Thus, the kinetic response of this formulation likely reflects the combined influence of diffusion, matrix hydration, and structural reorganization, even though the individual relaxational contribution could not be robustly separated by the model.

The behavior of CMC/U is also consistent with this coupled interpretation. Although it lacks CA-mediated crosslinking, the material undergoes rapid hydration and loss of structural integrity, forming a viscous and gelatinous phase during immersion. Consequently, its release behavior cannot be described as diffusion through a stable polymeric network alone, but rather as the combined effect of solute transport and changes in the physical state of the matrix. For CMC/U/BC/CA20 and CMC/U/BC/CA30, the lower swelling relative to CMC/U/BC/CA10 did not translate into slower urea release, reinforcing that transport is governed not only by water uptake but also by matrix organization, rigidity, heterogeneity, and the continuity of the available transport pathways.

The combined kinetic and swelling results indicate that urea release from the CMC-based composites involves an important diffusional contribution coupled with formulation-dependent polymer relaxation and structural changes. Therefore, the good agreement with the Higuchi model and the low Ritger–Peppas n values are interpreted as evidence of a substantial diffusional component rather than proof of a predominantly Fickian mechanism.

While the aqueous experiment provides an accelerated comparison of urea dissolution from the different matrices, the subsequent incubation experiment over a

total of 42 days provides complementary information on N transformations under solid-phase conditions.

The use of pure sand and incubation at 40 °C provided stringent experimental conditions for evaluating whether the composite matrices could influence N transformations and losses under conditions favorable to rapid nutrient mobility and transformation. Figure 8 shows urea mineralization, monitored through ammonia volatilization (Fig. 8A) and ammonium formation in soil (Fig. 8B). The experiments were performed at 40 °C, representing an accelerated condition that enhances urea hydrolysis and N losses, allowing a more rigorous evaluation of the systems.

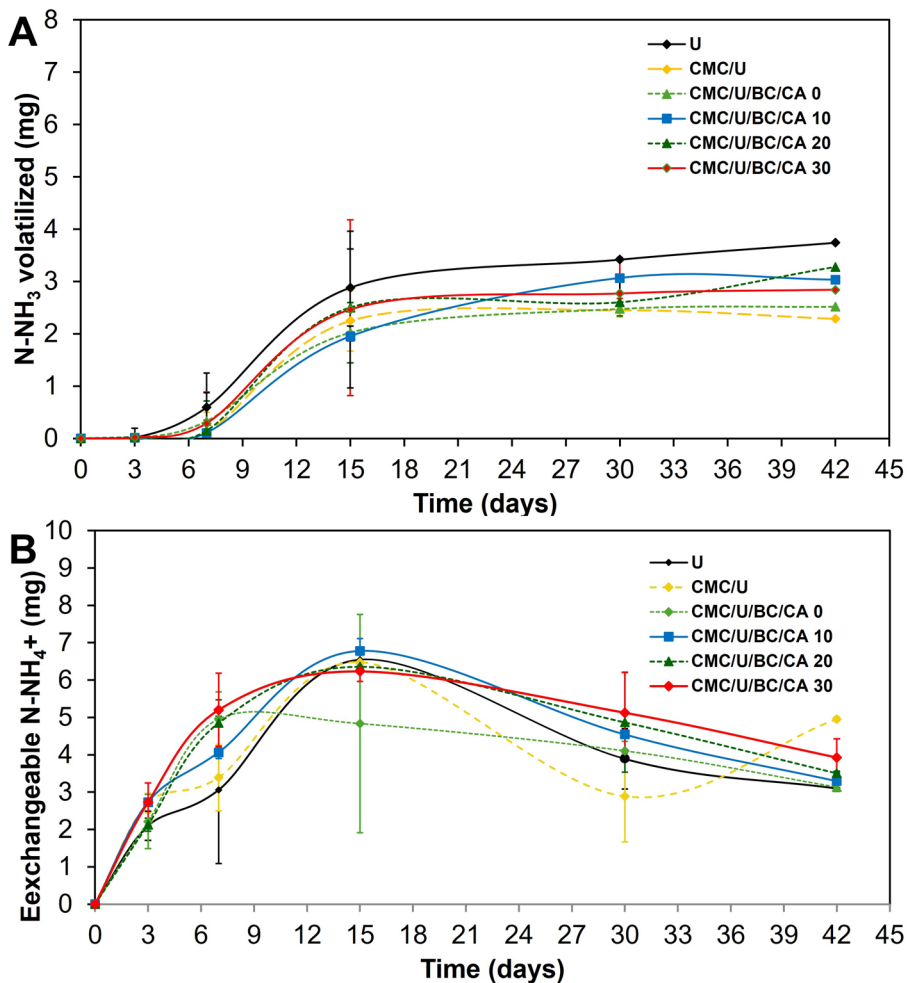


Fig. 8 Nitrogen transformation during the 42-day incubation experiment: **(A)** cumulative N volatilization as N-NH₃ and **(B)** exchangeable N-NH₄⁺ recovered from sand. Values are presented as mean±standard deviation (*n*=3). Different lowercase letters indicate significant differences among fertilizer treatments at each incubation time according to Tukey's HSD test (*p*<0.05)

As shown in Fig. 8A, all composite systems exhibited lower N-NH₃ volatilization compared to pure urea throughout the entire experimental period. This behavior is consistent across all time points. It is supported by the statistical analysis (Table S5), which shows that, despite slight variations, no significant differences were observed among the polymeric systems in most cases. In contrast, pure urea consistently presented the highest volatilization values, confirming the absence of any barrier to N loss.

However, after 42 days of incubation, cumulative N-NH₃ volatilization differed significantly among treatments (one-way ANOVA, $p < 0.001$; Table S6). Urea showed the highest cumulative N-NH₃ loss (3.74 ± 0.02 mg N), although it did not differ significantly from CMC/U/BC/CA 20 (3.27 ± 0.18 mg N). In contrast, CMC/U exhibited the lowest N-NH₃ volatilization (2.29 ± 0.13 mg N), representing a significant reduction compared with urea. CMC/U/BC/CA 0 and CMC/U/BC/CA 30 also showed significantly lower N-NH₃ volatilization than urea, whereas CMC/U/BC/CA10 exhibited an intermediate response.

A different response was observed for ammonium formation (Fig. 8B). At early times (up to 7 days), the composite systems exhibited higher exchangeable N-NH₄⁺ contents than pure urea, indicating that the formulations affected the distribution of N within the measured NH₄⁺ pool during the initial incubation period. After 14 days, the values became more similar across treatments, suggesting convergence in N transformation dynamics over time. Exchangeable N-NH₄⁺ also differed significantly among treatments at 42 days (one-way ANOVA, $p < 0.001$; Table S7). CMC/U showed the highest N-NH₄⁺ content (4.96 ± 0.06 mg N), significantly exceeding all other treatments. Among the biochar-containing composites, CMC/U/BC/CA 30 showed the highest numerical N-NH₄⁺ content (3.92 ± 0.50 mg N) and was significantly higher than urea, CMC/U/BC/CA 0, and CMC/U/BC/CA 10, but did not differ significantly from CMC/U/BC/CA 20. In contrast, CMC/U/BC/CA 0, CMC/U/BC/CA 10, and CMC/U/BC/CA20 did not differ significantly from conventional urea.

Regardless of the release behavior observed in aqueous medium, the composites generally resulted in higher exchangeable NH₄⁺ contents and lower measured NH₃ volatilization than pure urea. These results indicate that the polymeric formulations modified the distribution of N between the two pools quantified in this experiment. However, because N-NO₃⁻, residual matrix-associated N, and other potential N pools were not determined, these results cannot be interpreted as evidence of complete N retention or used to establish the overall fate of the applied N [43, 65]. Consequently, N remains available for a longer period, favoring its conversion to NH₄⁺ rather than its loss via volatilization.

To further assess the fate of N during incubation, a partial N recovery analysis was performed at 42 days by combining cumulative N-NH₃ volatilization and exchangeable N-NH₄⁺ (Table 2). Considering the initial application of 10 mg N, the measured recovery ranged from 56.4 to 72.4% and differed significantly among treatments (one-way ANOVA, $p = 0.0086$). CMC/U showed the highest measured partial N recovery (72.4%), resulting from 2.29 mg N recovered as volatilized NH₃ and 4.96 mg N as exchangeable NH₄⁺. This recovery was significantly higher than that observed for CMC/U/BC/CA0 (56.4%), which showed the lowest measured recovery. Pristine urea showed 68.4% partial N recovery and also differed significantly from CMC/U/

Table 2 Partial nitrogen recovery after 42 days of incubation, considering cumulative N-NH₃ volatilization and exchangeable N-NH₄⁺. Values are presented as mean ± standard deviation (n = 3). Different lowercase letters in the measured N recovery column indicate significant differences among treatments according to Tukey's HSD test (p < 0.05). Measured N recovery was calculated as the sum of cumulative N-NH₃ volatilization and exchangeable N-NH₄⁺ after 42 days relative to the 10 mg N initially applied. The unaccounted fraction represents N not recovered in these two measured pools; N-NO₃⁻ and residual N associated with the composite matrix were not directly quantified

Treatments	N-NH ₃ (mg)	N-NH ₄ ⁺	Measured N recovery	Recovery (%)	Unaccounted N
U	3.74 ± 0.02	3.10 ± 0.03	6.84 ± 0.02 a	68.4	31.6
CMC/U	2.29 ± 0.13	4.96 ± 0.06	7.24 ± 0.09 a	72.4	27.6
CMC/U/BC/CA0	2.51 ± 0.25	3.13 ± 0.12	5.64 ± 0.21 b	56.4	43.6
CMC/U/BC/CA10	3.03 ± 0.39	3.29 ± 0.22	6.33 ± 0.53 ab	63.3	36.7
CMC/U/BC/CA20	3.27 ± 0.18	3.50 ± 0.06	6.78 ± 0.16 ab	67.8	32.2
CMC/U/BC/CA30	2.84 ± 0.37	3.92 ± 0.50	6.76 ± 0.82 ab	67.6	32.4

B0. CMC/U/B10, CMC/U/B20, and CMC/U/B30 showed intermediate recoveries of 63.3, 67.8, and 67.6%, respectively, with no significant differences from either group according to Tukey's test ($p < 0.05$).

The remaining fraction, corresponding to 27.6–43.6% of the initially applied N, was therefore classified as unaccounted N rather than being assigned to a specific N pool. This fraction may include N-NO₃⁻, residual matrix-associated N, microbial immobilization, or other N pools that were not quantified in the present experiment. Because these fractions were not directly measured, their individual contributions cannot be determined.

Therefore, the present analysis represents a partial N recovery rather than a complete N balance. The results demonstrate that the formulations affected the distribution of N between the measured volatilized N-NH₃ and exchangeable N-NH₄⁺ pools, while 27.6–43.6% of the applied N remained unaccounted for. Direct quantification of N-NO₃⁻, residual matrix-associated N, and other relevant N pools would be required to establish a complete N balance.

Conclusions

The results demonstrate that CMC can be effectively used as a polymeric matrix for controlled urea release when combined with CA as a crosslinking agent and BC as a carbonaceous component. XRD, FTIR, DSC, and thermal analyses showed formulation-dependent changes in the crystalline organization, local environment, and thermal behavior of urea without evidence for the formation of a specific new crystalline phase. CA content strongly affected matrix hydration and structural stability, with CMC/U/BC/CA10 exhibiting the highest swelling while maintaining physical integrity, whereas the non-crosslinked systems progressively lost their structure during immersion. In aqueous medium, all CMC-based matrices delayed urea release relative to pure urea, with CMC/U/BC/CA10 showing the slowest release profile. Increasing CA content beyond 10 wt% did not proportionally improve release control, indicating that urea transport depended on the combined effects of hydration, matrix organization, rigidity, and heterogeneity. Peppas–Sahlin analysis further

showed that release involved coupled diffusional and polymer-relaxational contributions and therefore could not be described as purely Fickian. During the 42-day incubation experiment, the formulations significantly affected the measured N-NH_3 and exchangeable N-NH_4^+ pools. CMC/U showed the lowest cumulative N-NH_3 volatilization and the highest exchangeable N-NH_4^+ content at 42 days. Partial N recovery based on these two measured pools ranged from 56.4 to 72.4%, leaving 27.6–43.6% of the applied N unaccounted for. Because N-NO_3^- , residual matrix-associated N, and other N pools were not quantified, these results do not constitute a complete N balance and do not allow the fate of the unrecovered N to be established. Among the CA concentrations evaluated, 10 wt% provided the most favorable balance between swelling capacity, structural integrity, and controlled urea release, highlighting the importance of optimizing matrix architecture rather than simply increasing cross-linker content in the development of CMC-based controlled-release fertilizers.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00289-026-06685-y>.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Amanda S. Giroto¹ · Thaynara R. Prado¹ · Amanda Yashima¹ · Stella F. Valle² · Bruna Lendzion Alves³ · Maraisa Gonçalves¹

✉ Amanda S. Giroto
asgiroto@unifesp.br

¹ Science and Technology Institute, Federal University of São Paulo, Talim Street, 330, São José dos Campos 12231-280, SP, Brazil

² Lorena School of Engineering (EEL), University of São Paulo (USP), Estrada Municipal do Campinho, 100, Lorena 1260-281, SP, Brazil

³ Embrapa Instrumentation, XV de Novembro Street, 1452, São Carlos 13560-970, SP, Brazil